MTH636 – Game Theory

Lecture 21: Kuhn's Theorem & SPE Existence

1 Existence of Nash Equilibria and SPE in Finite Games

Theorem: Kuhn's Theorem

Every extensive form game with **perfect information** has at least one Nash Equilibrium in pure strategies.

Theorem: SPE Existence Theorem

Every extensive form game with **perfect information** has at least one Subgame Perfect Equilibrium (SPE) in pure strategies.

Note

This SPE can be obtained using the **backward induction principle** – i.e., the backward induction equilibrium is always a SPE.

2 Proof Sketch of SPE Existence via Backward Induction

The proof proceeds by constructing a strategy profile s^v via backward induction and then showing it constitutes a SPE. We outline the construction below.

Setup. Consider the game tree rooted at n_0. At the root, Nature (or a player) chooses from branches $p_1, p_2, \ldots, p_k$, leading to subtrees rooted at $n_1, n_2, \ldots, n_k$, respectively. For each subtree rooted at n_ℓ, we inductively obtain a strategy tuple:

$$s^{n_\ell} = (s_1^{n_\ell}, \ \ldots, \ s_n^{n_\ell})$$

where $s_i^{n_\ell}$ is player i's strategy within the subtree rooted at n_ℓ.

2.1 Construction of the Equilibrium Strategy

Case (i): $n_0 \in V_0$ (Nature's node).

For player i, define the strategy s_i^v as follows: *play* $s_i^{n_\ell}$ *if the game reaches* n_ℓ. Let

$$s^v = (s_1^v, \ \ldots, \ s_n^v).$$

We then show that s^v is a SPE.

Case (ii): $n_0 \in V_{i_0}$, $i_0 \in N$ (some player's node).

For player $i \neq i_0$: if the game reaches n_ℓ, play $s_i^{n_\ell}$. Call this strategy s_i^v.

For player $i = i_0$: at node n_0, choose n_{ℓ_0} such that

$$U_{i_0}(s^{x_{\ell_0}}) \geq U_{i_0}(s^{n_\ell}) \quad \text{for all } \ell \in \{1, \ldots, k\},$$

i.e., choose the branch n_{ℓ_0} that maximises player i_0's payoff. Then if the game goes to n_ℓ, follow strategy $s_{i_0}^{n_\ell}$.

Note

The exact (formal) proof can be found in **Solani's book**.

3 Worked Example: SPE via Backward Induction

Consider the following extensive form game. Nature (N) moves first at the root n_0, choosing Left (L) or Right (R). Player M (Man) then moves in each subtree.

Left subtree (after L): Man (M) moves, choosing F or D.

- F: Woman (W) moves, choosing F (payoff $(2, 1)$) or D (payoff $(0, 0)$).
- D: Woman (W) moves, choosing F (payoff $(0, 0)$) or D (payoff $(1, 2)$).

Right subtree (after R): Woman (W) moves, choosing F or D.

- F: Man (M) moves, choosing F (payoff $(2, 1)$) or D (payoff $(0, 0)$).
- D: Man (M) moves, choosing F (payoff $(0, 0)$) or D (payoff $(1, 2)$).

3.1 Backward Induction in the Left Subtree

Applying backward induction in the left subtree (subgame rooted after L):

- Woman's node after M plays F: Woman chooses F (payoff $1 > 0$). Resolved payoff: $(2, 1)$.
- Woman's node after M plays D: Woman chooses D (payoff $2 > 0$). Resolved payoff: $(1, 2)$.
- Man's node: compares $F \to (2, 1)$ and $D \to (1, 2)$. Man chooses F (payoff $2 > 1$).

The backward induction equilibrium within the left subgame is $\Big(F, (F, D)\Big)$ with payoff $(2, 1)$.

3.2 Backward Induction in the Right Subtree

Applying backward induction in the right subtree (subgame rooted after R):

- Man's node after W plays F: Man chooses F (payoff $2 > 0$). Resolved payoff: $(2, 1)$.
- Man's node after W plays D: Man chooses D (payoff $1 > 0$). Resolved payoff: $(1, 2)$.
- Woman's node: compares $F \to (2, 1)$ and $D \to (1, 2)$. Woman chooses D (payoff $2 > 1$).

The backward induction equilibrium within the right subgame is $\Big(D, (F, D)\Big)$ with payoff $(1, 2)$.

3.3 SPE for the Whole Game

Combining the backward induction outcomes of both subgames:

- Man's strategy: $M - (L,\ F,\ (F, D))$, i.e., play F in the left subgame and (F, D) in the right subgame.
- Woman's strategy: $W - (F, D,\ D)$, i.e., play (F, D) in the left subgame and D in the right subgame.

Concisely stated:

$$M:\ \Big(L,\ F,\ (F, D)\Big), \qquad W:\ (F, D,\ D)$$

forms the **SPE for the whole game**.

Note

Player M's expected utility: $U_M\Big((L, F, (F, D)),\ (F, D, D)\Big) = 2$.

4 Application: Chess and Determined Games

4.1 Structure of Chess as an Extensive Form Game

Chess is a two-player, perfect information, zero-sum game. The game tree has:

- A root node O.
- Subtrees for Player 1 (P-1), a "Draw" node (D), Player 2 (P-2), and Player 1 (P-1) again as the main branches.

The payoff matrix (from Player 1's perspective) takes the form:

	s_2^1	$\cdots$	s_h^2	
s_1^1	$(1, -1)$	$\cdots$		
s_2^1	$(0, 0)$	$\cdots$		
$\vdots$				
s_ℓ^1	$(1, -1)$	$(1_\ell = 1)$	$\cdots$	$(1, -1)$
$\vdots$				
s_m^1	$(-1, 1)$	$\cdots$		

Here the payoffs represent: $(1, -1)$ = Player 1 wins, $(0, 0)$ = Draw, $(-1, 1)$ = Player 2 wins.

4.2 Zermelo's Theorem Applied to Chess

By the SPE existence theorem (Kuhn's Theorem), chess – being a finite, two-player, perfect information game – must have a pure strategy SPE. Because the game is zero-sum, this means:

Theorem: Determinacy of Chess

Chess is a **determined** game: either Player 1 has a winning strategy, or Player 2 has a winning strategy, or both players can force a draw.

Formally, the minimax values satisfy:

$$\underline{v} = \bar{v} = 1 \quad \text{(for Chess, the value exists)}$$

where $\underline{v}$ is the maximin value and $\bar{v}$ is the minimax value. **For Chess**, we know:

$$\underline{v} = \bar{v} = 1$$

Note

While Zermelo's theorem guarantees that a winning (or drawing) strategy *exists* for one of the players, it does *not* tell us *which* player has a winning strategy or what that strategy is. Finding such a strategy for Chess remains an open computational problem.

Key Takeaway

The key takeaway of Lectures 20–21 is the refinement chain:

$$\text{SPE} \subset \text{Nash Equilibrium}$$

SPE eliminates non-credible threats by requiring rationality at *every* information set, not just along the equilibrium path. Backward induction provides a constructive method to find SPE in games with perfect information, and Kuhn's theorem guarantees such equilibria exist in pure strategies.